S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code: 0059)

DEPARTMENT OF COMPUTER ENGINEERING

DAILY DIARY

Seminar Presentation and Project Initiation (SPI)

Course Code: 315003

Name of Student	Mr. Shaikh Mohammad Zakariya Hafizuddin
Enrollment Number	24610960279
Exam Seat Number	388570
Roll Number	83
Class / Semester	CO5K (Third Year, Fifth Semester)
Academic Year	2026 – 2027
Seminar Topic	Smart Inventory Management System using IoT and AI/ML
Name of Guide	Prof. N. D. Patel

Group Members

Sr. No.	Enrollment No.	Seat No.	Name of Student
1	24610960279	388570	Shaikh Mohammad Zakariya Hafizuddin
2	24610960270	388571	Shaikh Suher Rayaz
3	24610960271	388572	Sharma Nilesh Ashok
4	24610960300	388599	Ansari Zeeshan Habib

PLAN OF WORK

Phase	Duration	Activity
1	07/09/2026 – 25/09/2026	Orientation, study of emerging technologies, topic selection
2	26/09/2026	Topic finalized: Smart Inventory Management using IoT and AI/ML
3	26/09/2026 – 08/10/2026	Searching, reading and comparing research papers (3 papers per member)
4	08/10/2026 – 16/10/2026	Preparation of SPI report and seminar presentation (PPT)
5	17/10/2026 – 19/10/2026	Corrections by course instructor, rehearsal and seminar presentation (19/10/2026)

My research papers (3 of the 12 papers reviewed by the group):

1. Singh, N. and Adhikari, D. (2023), “AI and IoT: A Future Perspective on Inventory Management,” IJRASET, 11(XI).

2. Reddy, B. N. et al. (2025), “Smart Inventory Management System,” IJIRSET, 14(4).

3. Chukomin, A. et al. (2023), “Development of an IoT-based Inventory Management Solution and Training Module Using Smart Bins,” CLF 2023.

My role in the group:

• Searched and studied 3 research papers on AI, IoT, smart inventory systems and smart bins.

• Compiled and formatted the SPI report, including literature review, comparison table, diagrams, project initiation and references.

• Prepared the seminar PPT and corrected it as suggested by the course instructor.

• Reviewed the report and PPT, added reference numbers and practised viva questions.

• Explained slides 13–14 during the seminar presentation, covering the problem statement, scope and objectives.

DAILY WORK RECORD

Sr. No.	Date / Day	Work Done / Activity	Outcome	Guide's Sign
1	07/09/2026 Monday	Attended the SPI course orientation. Learned the purpose of the seminar, project initiation, daily diary writing and evaluation process.	Understood SPI requirements	
2	09/09/2026 Wednesday	Formed the group with Nilesh, Zeeshan and Suher. Discussed our interests and planned to explore emerging-technology topics.	Group formed; topic search started	
3	11/09/2026 Friday	Read introductory material about AI/ML, IoT, blockchain, cloud computing and other emerging technologies.	Basic overview of emerging technologies	

Week 1 – Remarks of Guide: _____

Sr. No.	Date / Day	Work Done / Activity	Outcome	Guide's Sign
4	14/09/2026 Monday	Brainstormed real-life problem topics, including Brain–Computer Interface (BCI), air-pollution monitoring, smart waste management, smart traffic and smart inventory.	Potential topics shortlisted	
5	16/09/2026 Wednesday	Studied IoT basics: sensors, RFID, microcontrollers such as ESP32 and Raspberry Pi, and how devices transmit data.	Understood basic IoT workflow	
6	18/09/2026 Friday	Reviewed AI/ML basics, including classification, regression and how prediction can help estimate future inventory demand.	Basic understanding of AI/ML	

Week 2 – Remarks of Guide: _____

Sr. No.	Date / Day	Work Done / Activity	Outcome	Guide's Sign
7	21/09/2026 Monday	Compared BCI, air-pollution monitoring, smart waste, smart traffic and smart inventory by usefulness, available research and project feasibility.	Topics compared for feasibility	
8	23/09/2026 Wednesday	Investigated manual stock-register and physical-counting problems, such as stock-outs, over-purchasing, missing items and limited usage history.	Inventory problem identified	
9	25/09/2026 Friday	Prepared a short proposal after discussing the shortlisted ideas and their practical feasibility with the group.	Topic proposal discussed	
10	26/09/2026 Saturday	Finalized "Smart Inventory Management System using IoT and AI/ML" as the seminar topic.	Seminar topic finalized	

Week 3 – Remarks of Guide: _____

Sr. No.	Date / Day	Work Done / Activity	Outcome	Guide's Sign
11	28/09/2026 Monday	Practised searching for research papers and used keywords related to AI, IoT, RFID, smart bins and inventory forecasting.	Research-paper search started	
12	29/09/2026 Tuesday	Organized the group research work and selected papers covering AI with IoT, smart inventory software and IoT-based smart bins.	Paper-reading plan prepared	
13	30/09/2026 Wednesday	Read "AI and IoT: A Future Perspective on Inventory Management" and noted how IoT data and AI can support real-time tracking and prediction.	Research Paper 1 reviewed	
14	01/10/2026 Thursday	Made notes on Paper 1 about real-time inventory visibility, predictive analytics, automated replenishment and challenges such as data security and sensor reliability.	Key findings and limitations noted	
15	03/10/2026 Saturday	Read "Smart Inventory Management System" and studied its web-based implementation using Node.js, Express.js, EJS, MySQL and Bcrypt.js.	Research Paper 2 reviewed	

Week 4 – Remarks of Guide: _____

Sr. No.	Date / Day	Work Done / Activity	Outcome	Guide's Sign
16	05/10/2026 Monday	Read the paper on an IoT-based inventory solution using smart bins. Noted weight sensors, IoT Hub, AI analysis and the Theory of Constraints.	Research Paper 3 reviewed	
17	06/10/2026 Tuesday	Discussed the papers with the group and organized their objectives, technologies, approaches and future-scope points.	Research findings organized	
18	07/10/2026 Wednesday	Compared the papers by tracking method, software technologies, use of AI, benefits and limitations; noted possible future extensions.	Literature comparison prepared	
19	08/10/2026 Thursday	Identified the gap between basic stock tracking and demand prediction, along with cost, sensor reliability and security considerations.	Research gap identified	

Week 5 – Remarks of Guide: _____

Sr. No.	Date / Day	Work Done / Activity	Outcome	Guide's Sign
20	09/10/2026 Friday	Reviewed the college SPI report format and arranged the planned chapters, headings, diagrams and reference style.	Report structure planned	
21	10/10/2026 Saturday	Drafted the Abstract and Introduction, explaining the inventory problem and the role of IoT and AI/ML.	Abstract and Introduction drafted	
22	12/10/2026 Monday	Prepared the Literature Review and a comparison table summarizing the selected research papers.	Literature Review drafted	
23	13/10/2026 Tuesday	Wrote the information chapter about IoT-based stock tracking and AI/ML forecasting; prepared the system flowchart and architecture diagram.	Technical chapter and diagrams prepared	
24	14/10/2026 Wednesday	Wrote the advantages, limitations, conclusion and project-initiation details, including problem statement, scope and objectives.	Main report chapters drafted	
25	15/10/2026 Thursday	Added the bibliography and references, then prepared the seminar PPT using the college format.	References added; PPT prepared	
26	16/10/2026 Friday	Reviewed the report and PPT with the group, corrected the content and layout, and printed and submitted the documents.	Report and PPT submitted	

Week 6 – Remarks of Guide: _____

Sr. No.	Date / Day	Work Done / Activity	Outcome	Guide's Sign
27	17/10/2026 Saturday	Applied the course instructor's corrections to formatting, introduction details and research-paper references.	Instructor's corrections completed	
28	18/10/2026 Sunday	Checked report page numbers and PPT references, rehearsed the presentation and revised likely viva questions.	Presentation and viva preparation completed	
29	19/10/2026 Monday	Presented the seminar and explained slides 13 and 14, covering the problem statement, scope and objectives of the Smart Inventory Management System.	Slides 13–14 explained in seminar	

Week 7 – Remarks of Guide: _____

LEARNING OUTCOMES

• Learned how to select a seminar topic from emerging technologies based on a real-life problem.

• Learned to search, read and compare research papers using Google Scholar, IEEE Xplore and ResearchGate.

• Understood how IoT sensors and smart bins track inventory and send real-time data.

• Learned how AI/ML can analyse stock data and support demand forecasting.

• Understood how bottlenecks and stock replenishment can be improved using the Theory of Constraints.

• Learned to prepare a technical report with diagrams, tables and references.

• Improved presentation and communication skills by explaining slides 13–14 and answering viva questions.

Other learning: Smart bins use weight sensors to measure stock; IoT collects data, while AI and the Theory of Constraints help analyse inventory and improve replenishment.

Signature of Student

Name: Shaikh Mohammad Zakariya Hafizuddin

Date: _____

Signature of Guide

Name: Prof. N. D. Patel

Date: _____

Signature of Head of Department: _____